

ABHISHEK PARITA

abhishek13parita25@gmail.com · [LinkedIn](#) · [GitHub](#)

EXPERIENCE

AIML Developer Intern · Advantal Technology, Indore

Nov 2025 – Feb 2026

- Built a **real-time BGP anomaly detection system** using ensemble ML (LSTM Autoencoder + Isolation Forest + Heuristics) on sequential network data, replacing a single-model approach and improving detection coverage.
- Designed **feature-level drift monitoring** (KS-test) comparing live vs. training distributions; triggered automated retraining alerts before production model degradation.
- Delivered a **modular, retraining-ready ML pipeline** — training → validation → anomaly scoring → performance tracking — reducing manual monitoring overhead.

PROJECTS

CrackIt — AI Interview Prep Platform

Apr 2026

- Built **FastAPI backend** with **SQLAlchemy/SQLite** ORM, REST endpoints, and Pydantic schema validation; consumed by a React/Vite frontend.
- Engineered end-to-end **RAG pipeline**: LangChain + ChromaDB for vector storage, HuggingFace Sentence Transformers (`all-MiniLM-L6-v2`) for embeddings, context-injected prompting to ground **Llama 3.2** responses and eliminate hallucination.
- Designed full system from API contracts to inference layer — demonstrates ownership across backend, ML integration, and data flow.

Large-Scale Financial Fraud Detection

Sept 2025

- Trained **XGBoost classifier** on 6.3M+ transactions (0.13% fraud rate); tackled extreme class imbalance via cost-sensitive learning and decision-threshold optimisation.
- Achieved **98.3% Recall** and **PR-AUC 0.865** — metrics directly aligned with minimising missed fraud in production cost-sensitive scenarios.

ML Model Monitoring & Drift Detection System

Jan 2026

- Built **production monitoring framework**: FastAPI REST API, SQLite prediction logging, severity-based alerting schema.
- Implemented **statistical drift detection** (KS-test + Chi-Square) with interactive dashboard surfacing drift metrics and confidence trends in real time.

Credit Scoring System with Explainability

Feb 2026

- XGBoost credit-default predictor with engineered behavioural features recall-tuned via class weighting for high-risk applicant detection.
- Applied **SHAP** for global feature importance and per-prediction explanations — audit-ready for fintech compliance and risk review workflows.

TECHNICAL SKILLS

Languages: Python, SQL

ML / AI: XGBoost, Scikit-learn, LSTM Autoencoder, Isolation Forest, SHAP, Anomaly Detection, Drift Detection (KS-test, Chi-Square), Imbalanced Classification

LLMs & RAG: LangChain, ChromaDB, HuggingFace Inference API, Sentence Transformers, RAG Pipelines

Backend & APIs: FastAPI, SQLAlchemy, REST API Design, Pydantic, Uvicorn

Data & Tools: Pandas, NumPy, Matplotlib, Jupyter, Github

EDUCATION

B.Tech – Artificial Intelligence & Machine Learning

Sept 2022 – Present

Indore Institute of Science and Technology, Indore